INDEPENDENT METHODOLOGICAL VERIFICATION REPORT

Auditing Measurement Validity in Campaign Classification on LEN-Small

A Rigorous Evaluation of Size Confounding, Nearest-Neighbor Matching, and Restricted-Observation Transferability

Document: reports/methodology_audit.md	**Target Dataset:** LEN-Small (103 Canonical Events)	**Validation Status: VERIFIED & REPRODUCIBLE**
Protocol: 5-Fold Stratified Cross-Validation	**Evaluated Models:** LogReg (L2), HGBT Trees	**Audit Date:** October 2026

1. Executive Summary & Audit Verdict

This independent methodological audit verifies the experimental pipeline, mathematical claims, data handling integrity, and statistical boundaries of the study *'Beyond Graph Scale: Auditing Size Confounding and Retained-Observation Robustness in Campaign Classification on LEN-Small'*.

The core research finding is confirmed: **unconstrained campaign classification on LEN-Small is predominantly driven by graph scale** (log node count alone yields 0.8739 Balanced Accuracy, 95% CI: [0.805, 0.932]). However, size matching and observation budget restrictions reveal critical nuances that require precise methodological demarcation.

Dimension	Initial Guide Assumption	Verified Empirical Reality	Audit Verdict
Archive Format	ZIP archive with edgelists + CSV	Single tar archive (small_encoder_final.tar) with 104 JSONs.	**CORRECTED**
Cohort Size & Labels	100 events, standalone CSV	103 canonical events (1 duplicate removed via SHA-256). Labels in filenames.	**VERIFIED**
Size Distribution	Campaign assumed larger than comparison	Inverted: Comparison is ~5x larger (median 3,227 vs 506 nodes).	**AUDITED**
Scale Baseline	Assumed secondary to topology	Dominant: Log node count achieves 0.8739 BA (95% CI [0.805, 0.932]).	**VERIFIED**
Size Matching	Caliper matching on log node count	16 pairs (n=32); |SMD| drops from 2.24 to 0.10 (signed: -2.24 to -0.10). Non-scale features retain 0.750 BA.	**VERIFIED***
Edge Budget (K=2000)	Feasible across all K in {250,500,1000,2000}	Infeasible: Only 4 campaign graphs have >=2000 edges. 5-fold CV invalid.	**EXCLUDED**
D1 vs D2 Observation	Retrained models called 'approximations'	Fold-aligned frozen structural models (D1: 0.598 / 0.725 / 0.746 BA) vs retrained all-feature models (D2: 0.941 / 0.938 / 0.906). Not like-for-like.	**CORRECTED**

2. Ingestion, Duplicate Handling & Leakage Prevention

A. Canonical Deduplication: During archive extraction, an exact binary duplicate was identified: #SesimiziDuyanVarM[dotless i]___2023-03-23_campaign_fulldata.json and its ASCII equivalent ...VarMi... had identical SHA-256 hash (57da6c194213...). Ingestion script src/ingest_len.py logged the duplicate and retained exactly one canonical entry, establishing 103 verified event graphs.

B. Feature Leakage Prevention: Heavy 772-dim node_attr and 776-dim edge_attr continuous embeddings in the raw JSON files were discarded during ingestion. Feature matrices contain purely topological properties and aggregated text dispersion metrics. Event IDs, labels, fold IDs, and account usernames are strictly excluded from input tensors.

C. Partitioning & Inner Tuning: 5-fold cross-validation is partitioned strictly at the event level. Hyperparameter tuning for Logistic Regression regularization (C in {0.01, 0.1, 1.0, 10.0}) is performed via an inner 3-fold CV loop on the training fold

only, avoiding optimistic bias.

3. Size Matching: Calibration & Statistical Limits

To evaluate whether non-scale topological indicators (degree Gini, reciprocity, coreness) possess predictive utility beyond graph size, 1:1 nearest-neighbor matching without replacement was performed on log(1 + N) with a caliper of 0.25 x SD (0.2859).

Verification Results: The signed standardised mean difference (campaign minus comparison, pooled SD) moved from **-2.2374** to **-0.0954**, i.e. |SMD| 2.24 to 0.10, across 16 matched pairs (32 total events). On this balanced cohort:

- The scale model (log N + log E) drops from 0.8837 BA on the full cohort down to **0.5938 BA** (95% CI: [0.417, 0.753]), statistically overlapping random guessing (0.50).
- Structural features retain **0.6875 BA** (95% CI: [0.524, 0.857]), and content features retain **0.7500 BA** (95% CI: [0.598, 0.906]); these are exploratory results on 32 events.
- *Statistical Limitation:* Because N = 32, bootstrap 95% confidence intervals are broad (about +/-0.15). Matching acts as a diagnostic sensitivity stress-test rather than a definitive causal proof.

4. Observation-Budget Robustness & Infeasibility at K=2000

Observation-budget sampling evaluated classifier resilience when only K in {250, 500, 1000} edges are observed:

- **D2 (Retrained Models):** Classifiers trained and tested on sparse views remain accurate (BA >= 0.90 at every feasible budget). They use all 12 features recomputed on each sampled view.
- **D1 (Frozen Full-Graph Models):** The structural-group models from Exp A (7 features) are applied without retraining; each event is scored only by the fold model that held it out. BA: K=250: **0.5980** [0.505, 0.685] vs 0.7843 on complete graphs; K=500: **0.7246** [0.639, 0.807] vs 0.7821 on complete graphs; K=1000: **0.7456** [0.650, 0.842] vs 0.7567 on complete graphs. D1 and D2 use different feature sets, so their gap is not a controlled estimate of retraining. An earlier misaligned evaluation (superseded values 0.7558 and 0.8282 at K=500/1000) has been corrected.
- **K=2000 Infeasibility:** Only 4 campaign graphs have >= 2,000 edges. Partitioning 4 positive events into 5 cross-validation folds is mathematically impossible without producing empty positive test strata. K = 2000 was excluded from valid benchmark evaluation to avoid reporting distorted, unstratified metrics.

5. Benchmark Metrics Verification Matrix

Experiment ID	Feature Set / Protocol	N	Bal. Acc (95% CI)	Macro-F1	ROC-AUC
B0a_log_node_logreg	Scale: log N (full cohort)	103	0.8739 [0.805, 0.932]	0.8738	0.9340
B0b_log_node_edge_logreg	Scale: log N + log E (full cohort)	103	0.8837 [0.817, 0.941]	0.8835	0.9306
A_combined_logreg	All 12 features (full cohort)	103	0.9229 [0.867, 0.971]	0.9221	0.9857
C_matched_scale_logreg	Scale: log N + log E (matched)	32	0.5938 [0.417, 0.753]	0.5934	0.5703
C_matched_structure_logreg	Structure (matched)	32	0.6875 [0.524, 0.857]	0.6875	0.7227
C_matched_content_logreg	Content (matched)	32	0.7500 [0.598, 0.906]	0.7490	0.7734
D2_K250	D2 retrained, K=250 (12 features)	102	0.9412 [0.894, 0.984]	0.9412	0.9777
D2_K500	D2 retrained, K=500 (12 features)	98	0.9379 [0.888, 0.980]	0.9385	0.9538
D2_K1000	D2 retrained, K=1000 (12 features)	73	0.9063 [0.829, 0.970]	0.9118	0.9294
D1_K250_frozen	D1 frozen, K=250 (7 structural)	102	0.5980 [0.505, 0.685]	0.5866	0.7670
D1_K500_frozen	D1 frozen, K=500 (7 structural)	98	0.7246 [0.639, 0.807]	0.7245	0.8154
D1_K1000_frozen	D1 frozen, K=1000 (7 structural)	73	0.7456 [0.650, 0.842]	0.7591	0.8000

6. Audit Sign-Off & Methodological Verdict

VERDICT: REPRODUCIBLE, WITH BOUNDED CLAIMS. The pipeline is deterministic when seeded. A fold-alignment error in the frozen-model (D1) evaluation was found during the audit and corrected; the values above are the corrected results. Remaining limitations: 103 events (32 after matching); percentile-bootstrap intervals over fixed out-of-fold predictions that ignore refitting, fold-split, matched-pair and shared-account dependence; an unexplained difference between the analysed archive (104 files) and the documented LEN-Small release (100 graphs); and temporal features excluded because timestamp semantics are unverified.